CS23431 - OPERATING SYSTEM

EXP 8: First Fit Contiguous Memory Allocation Using C

QUESTION :

Write a C program to implement the First Fit contiguous memory allocation technique. The program should take the sizes of various memory blocks and the sizes of processes as input. For each process, search through the memory blocks sequentially and allocate the first block that meets the size requirement. If no such block exists, indicate that the process cannot be allocated. Finally, display the allocation details and calculate internal fragmentation.

CODE:

```
#include<stdio.h>
int main(){
 int m,n;
 printf("Enter number of memory blocks:\n");
 scanf("%d",&m);
 int memory[m],original[m];
 printf("Enter size of each block:\n");
 for(int i=0;i<m;i++){
   scanf("%d",&memory[i]);
   original[i]=memory[i];
 }
 printf("Enter number of processes:\n");
 scanf("%d",&n);
 int process[n];
 printf("Enter size of each process:\n");
 for(int i=0;i<n;i++){
   scanf("%d",&process[i]);
 }
 int count=0;
 while(count<n){
   int flag=0;
   for(int i=0;i<m;i++){
     if(process[count]<=memory[i]){
       printf("Process %d of size %d is allocated to Block %d of
size %d with Fragment
%d\n",count+1,process[count],i+1,original[i],(memory[i]-
process[count]));
       flag=1;
       memory[i]-=process[count];
       break;
```

```
        }
      }
      if(flag==0){
        printf("Process %d of size %d is not
allocated\n",count+1,process[count]);
      }
      count+=1;
    }
  }
```

Output:

Input :

```
5
100 500 200 300 600
4
212 417 112 426
```

Expected Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 2 of size 500
with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600
with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500
with Fragment 176
Process 4 of size 426 is not allocated
```

Output :

```
Enter number of memory blocks:
Enter size of each block:
Enter number of processes:
Enter size of each process:
Process 1 of size 212 is allocated to Block 2 of size 500 with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with Fragment 176
Process 4 of size 426 is not allocated
```